

AI Audit Report

1. Declaration and controls

OpenAI Codex was used on 17 August 2026 for specification interpretation, JMeter-plan generation and review, raw-JTL analysis, evidence traceability, and report drafting. Local execution, screenshots, and external publication remained student-controlled. AI-generated text was reviewed against source code, JMX, screenshots, and raw JTL logs before inclusion. No AI-generated performance result, screenshot, GitHub URL, or video URL is treated as evidence.

2. Material interaction record

Local date / stage	Tool	Student prompt or instruction	AI output	Human validation and outcome
17 Aug - scope	OpenAI Codex	Analyse the HW05 specification and select endpoint groups/FR traceability using prior HW03/HW04 work.	Proposed login, product detail, cart, and checkout; classified auth/read/transactional groups.	Checked against backend routes and retained FR-02, FR-06, FR-07, FR-08 only.
17 Aug - plan design	OpenAI Codex	Use JMeter and create equivalent Load, Stress, Spike plans with CSV data.	Generated JMX scaffolds with CSV variables, token extraction, assertions, and unique listener views.	Real preflight exposed a CSV/header mismatch, requiring execution review.
17 Aug - correction	OpenAI Codex	Diagnose login failures in the preflight.	Identified the CSV data-file/header contract as the likely cause.	Header was removed; the failed preflight was retained separately. Final Load, Stress, Spike, and Endurance JTLs ran successfully.
17 Aug - evidence	OpenAI Codex	Explain and organise JMeter/Task Manager, dxdiag, lockout, and reset evidence.	Supplied an evidence taxonomy and capture guidance.	EV-RUN, EV-HW, and EV-LOCKOUT artifacts were captured and verified on disk.
17 Aug - lockout review	OpenAI Codex	Verify lockout runtime before calling it a bug.	Required a valid-login check after two invalid requests rather than relying only on source.	Fresh runtime sequence was 401, 401, 403 for a valid password; the issue packet records the evidence.
17 Aug - result analysis	OpenAI Codex	Summarise raw JTL files without inference.	Used the submitted skill script to calculate samples, errors, mean, p95, duration, and RPS.	Main-report values match script output: Load 200; Stress 800; Spike 500; Endurance 4,400; all zero errors.
17 Aug - reporting	OpenAI Codex	Prepare professional reports, critique, skill mirror, and continuous-testing proposal.	Drafted traceable Markdown materials and PDF-rendering automation.	Main Report and AI Audit Report PDFs were rendered successfully; content was reviewed against the registered evidence.

3. AI misinterpretation hunt

AI-proposed claim or omission	Correct evidence	Why it was wrong or incomplete	Effect of review
A CSV header could coexist with explicit JMeter variable names.	Preflight JTL: 75 failed samples; final headerless CSV: successful runs.	JMeter used the first row as data when explicit variable names were supplied.	Preserved the failure for audit and excluded it from final metrics.
Lockout could be labelled a defect from source inspection alone.	Controlled runtime sequence after fresh start: 401, 401, then valid-password 403.	Source explains a hypothesis but does not prove user-visible behavior.	Performed the additional valid-login check before writing the issue packet.
Task Manager screenshot can establish a system maximum.	EV-HW-002 is a point-in-time monitor frame only.	It has no per-process, continuous peak measurement.	Report states a measured throughput lower bound and avoids a fabricated capacity ceiling.

4. Usage conclusion

AI accelerated design and documentation but did not replace testing. The acceptance source for performance claims is the JTL plus its execution configuration; screenshots establish attribution, not missing metrics. The completed critique in `ai_critique.md` records the resulting collaboration principle.